MAE339 – Aerospace Engineering Lab

Lab 3 – Heat Transfer In A Cylindrical Fin Under Steady State

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Abstract

The main purpose of this experiment was to determine the theoretical fin efficiency, effectiveness, and total heat rate based on the experimental data collected using FLIR E60 IR Camera & thermocouples sensors along with Research IR Software.
This was accomplished by observing the system under a steady state and achieving a percentage error of about 38% according to the MATLAB plots generated using the data collected during the physical experiment.

Introduction

The lab was about deducing the theoretical fin efficiency, effectiveness, total heat rate, Rayleigh and Nusselt numbers based on the experimental data collected using FLIR E60 IR Camera & thermocouples sensors along with Research IR Software.

Experimental Setup

FLIR E60 IR Camera

- The camera was set up on tripod which was not moved throughout the experiment.
- The equipment enabled us to get pixel data for processing using the Research IR software installed on the team laptop.

Thermocouples

- Thermocouples allowed us to verify whether IR camera was calibrated properly.
- They also enabled us to check the status of the IR camera when the experiment was in progress as we could cross check the approximate values in real time.

Anemometer

- It was used to calculate average windspeed for theoretical calculations.

Heating plate and cylindrical fin

- It was used to increase the temperature of the system until it reached steady state.
- The cylindrical fin was the main object under observation which the IR camera was pointed directly towards.

Fan

- It was used to introduce a constant average windspeed to the system for heat dissipation until steady state was achieved

Data Analysis

The data was analyzed using the equations provided in the lab manual and certain criteria were needed to be confirmed visually using the IR Camera snapshots i.e. confirming steady state and initial state of the system.

The snapshots, results from hand solving equations, and pixel data were used in MATLAB for drawing a conclusion and estimate error. Everything used for procession is as follows:

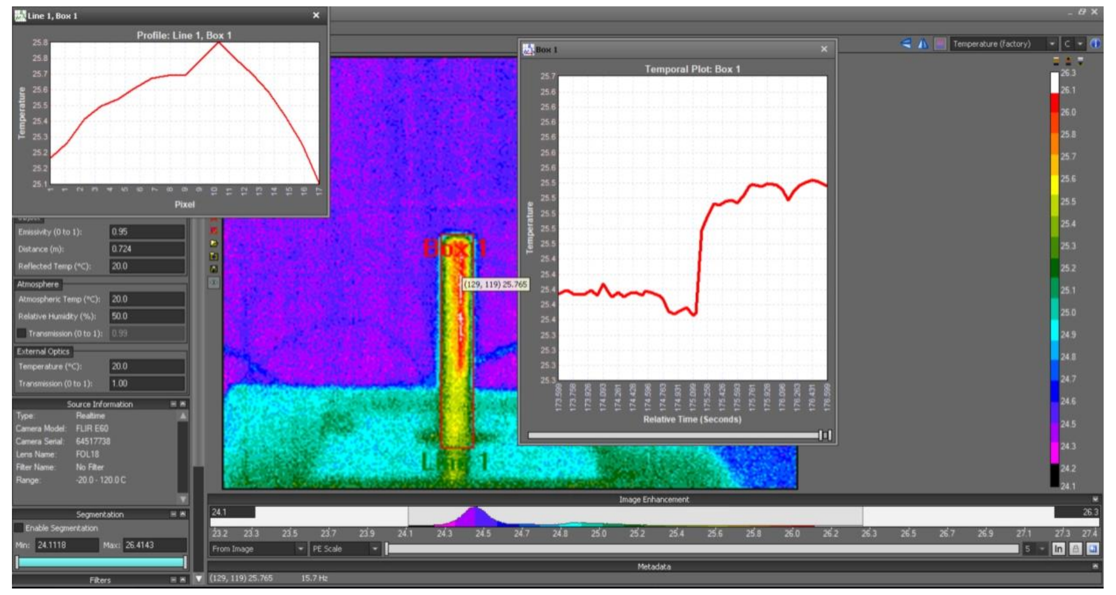


Figure A: Initial IR Image of the system along with the temporal plot

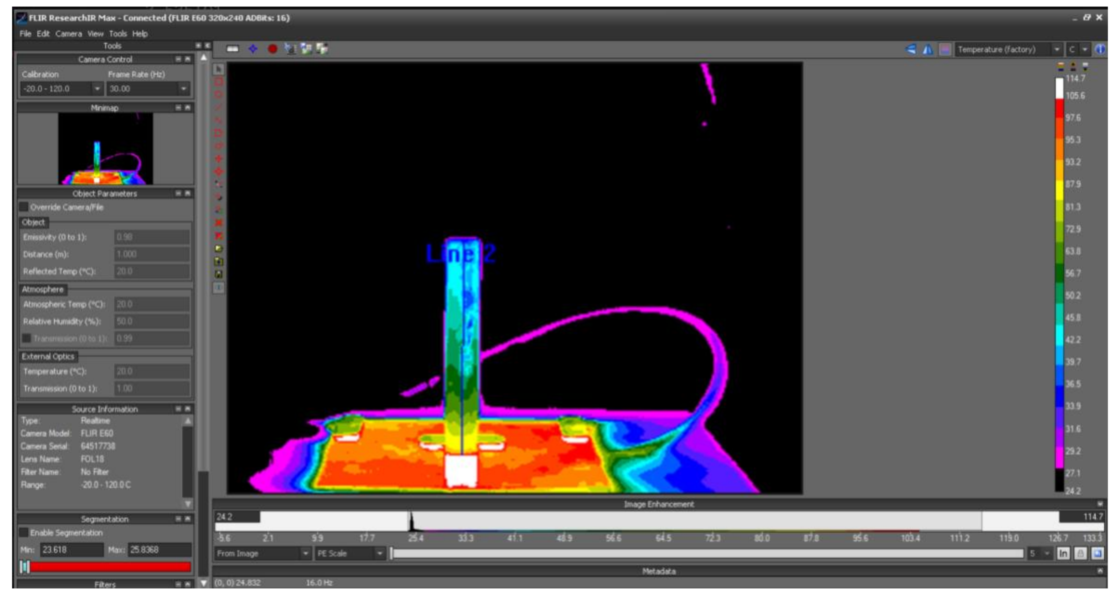


Figure B: IR Image of the system under steady state

HANDWORK AND EQUATIONS CONTINUED →

→ Part 1

$$U_\infty = \frac{2.4 + 2.4 + 1.5}{3} = 2.1\ m/s.$$

Diameter = 0.0192 m

$$Re_D = \frac{U_\infty \cdot D}{\nu} = \frac{2.1 \times 0.01912}{1.59 \times 10^{-5}} = \boxed{2525.283019.}$$

$$\overline{Nu_D} = 0.3 + \frac{0.62 \cdot Re_D^{1/2} \cdot Pr^{1/3}}{\left[1 + \left(\frac{0.4}{Pr}\right)^{2/3}\right]^{1/4}} \cdot \left[1 + \left(\frac{Re_D}{282{,}000}\right)^{5/8}\right]^{4/5}$$

$$= 0.3 + \frac{\left(0.62 \cdot \sqrt{Re_D} \cdot \sqrt[3]{0.707}\right)}{\left[1 + \left(\frac{0.4}{0.707}\right)^{2/3}\right]^{1/4}} \cdot \left[1 + \left(\frac{Re_D}{282{,}000}\right)^{5/8}\right]^{4/5}$$

$$\boxed{\overline{Nu_D} = 25.68263447}$$

$$Ra_{L_c} = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu\alpha}$$ (Using 'middle' IR Temp value)

$$Ra_{L_c} = \frac{9.18\left(\frac{1}{T_f}\right)(T_s - T_\infty)(0.12052)^3}{\nu\alpha}$$

$$T_f = \frac{95.3 + 24.3}{2} = 59.8$$

$$= \frac{9.18\left(\frac{1}{332.8}\right)(95.3 - 24.3)(0.12052)^3}{(1.59\times10^{-5})(2.25\times10^{-5})}$$

$$\boxed{Ra_{L_c} = 9583307.837}$$ (Along the length of the cylinder).

$$\overline{Nu}_{L_c} = 0.68 + \frac{0.67\, Ra_L^{1/4}}{\left[1 + \left(\frac{0.492}{Pr}\right)^{9/16}\right]^{4/9}}$$

$$\boxed{\overline{Nu}_{L_c} = 29.27845839.}$$

$$\boxed{\overline{Nu}_{Tot} = Nu_L + Nu_D = 54.96109283}$$

$$\boxed{h_c = \frac{\overline{Nu}_{Tot} \cdot k_f}{L_c} = 76.17515481}$$

→ Part 2

$$h_r = \varepsilon\sigma(4T_\infty^3) = (2.25\times10^{-5}) \times (5.67\times10^{-8}) \times 4 \times (297.45)^3$$

$$\boxed{h_r = 1.342973639 \times 10^{-4}}$$

$$h_{tot} = h_c + h_r$$

$$h_{tot} = 76.17528911$$

$$m = \sqrt{\left(\frac{97.41781356 \times (4\pi R + 2H)}{(50) \times (\pi (\frac{D}{2})^2)}\right)}$$

$$m = \sqrt{1916.44135315} \Rightarrow m = 43.77717764512$$

$$q_f = M \tanh(mL')$$

$$L + D/4 = 0.1253 = L'$$

$$\frac{\theta}{\theta_B} = \frac{\cosh[m(L' - x)]}{\cosh[mL']}$$

$$\theta_b = T_B - T_\infty$$

$$M = \theta_b (hPkA_c)^{1/2}.$$

$$\theta_b = 95.3 - 24.3 = 71\ \text{K and } C^\circ$$

$$\boxed{M = 44.6212546046}$$

$$q_f = M \tanh(mL')$$

$$\boxed{q_f = 44.61971959981}$$

$$\varepsilon_f = \frac{q_f}{hA_c\theta_b} = \frac{50.4603612658}{(97.\ldots)\left(\pi\left(\frac{19.12}{2000}\right)^2\right)(71)}$$

$$\boxed{\varepsilon_f = 28.7335125295}$$

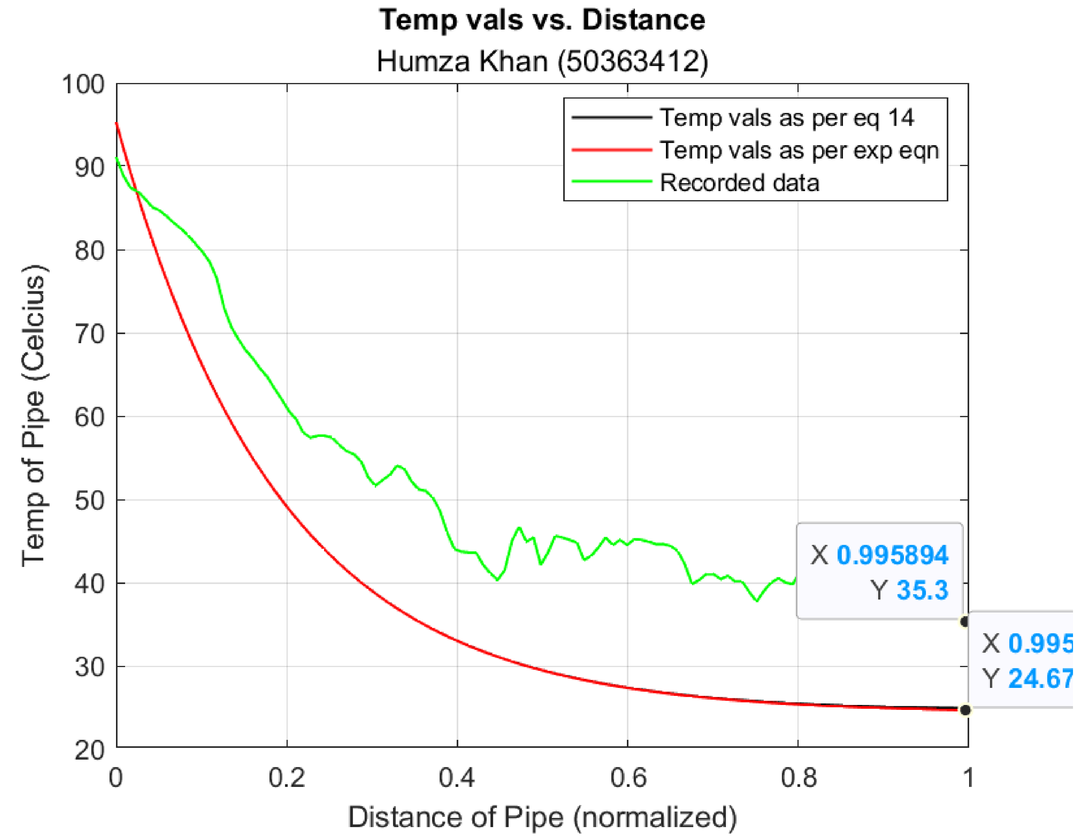
$$S_A = 7.81355 \times 10^{-3}$$

$$\eta_f = \frac{q_f}{hA_f\theta_b}$$

$$0.933695429878$$

→ HANDWORK AND EQUATIONS END HERE

Results (according to MATLAB and Hand Calculations)



- ε of the fin is about **28.7335125295** (should not be greater than 1, I can't figure out where I'm missing the factors.)
- q_f is about **44.6197195981W**
- n_f is about **0.933695429878** (is not exceeding 1, checks out)
- Predicted tip temperature of about **24.6 degrees Celsius** (which again seems to be off vs the actual value)

Conclusion

Finally I can conclude that the theoretical equations plot should be very close to the experimental values recorded if compared to previous similar experiments. However, with the data I got, and the mathematics done during data analysis, I conclude a percentage error of about **43%** according to my MATLAB & hand calculations.

There were some points that I was confused about, which may have contributed to such a large percentage error. For example, I may have done the data normalization incorrectly in the post processing.

There were more things done in the MATLAB code, such as offsetting the data by the ambient room temperature which made the curves fit better, but also reduced the overall percentage error in the values. Doing the offset makes sense as the temperatures at the base of the cylinder should have similar experimental and theoretical values. (which theta_B was not doing as expected)

The resulting plot and calculated values are posted in the previous section.